Uniform speed

If a body in motion covers equal distances in equal intervals of time, then the body is said to be in uniform speed.

Non- uniform speed

If a body covers unequal distances in equal intervals of time, the body is said to be in non-uniform speed.

$$\text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Time taken to travel the distance}}$$

1 km/h = 5/18 m/s

How we got this ?

1 km = 1000 m; 1 h = 3600 s

1 km / h = 1000 m / 3600 s = 5/ 18 m /s

Know the speed

Tortoise	0.1 m/s
Person walking	1.4 m / s
Falling raindrop	9-10 m / s
Cat running	14 m/s
Cycling	20-25 km/h
Cheetah running	31 m/s
Bowling speed of fast bowlers	90-100 miles /h
Badminton smash	80-90 m/s
Passenger jet	180 m/s

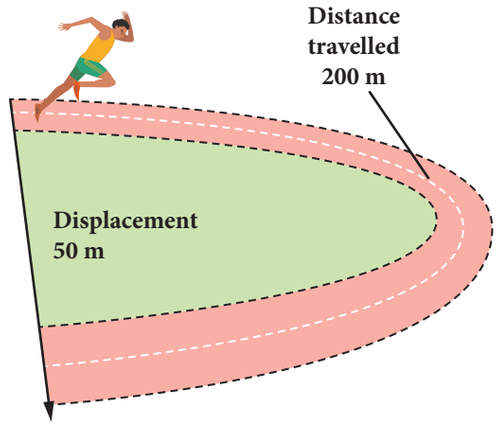
2.2.2 Velocity

Velocity is the rate of change in displacement.

Velocity (v) = Displacement / Time

SI unit of velocity is metre / second (m/s).

Look at the figure. An athlete takes 25 s to complete a 200 m sprint event. Find her speed and velocity.



$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{200}{25} = 8 \text{ m/s}$$

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{50}{25} = 2 \text{ m/s}$$

Uniform velocity

A body is said to have uniform velocity, if it covers equal displacement at equal intervals of time in the same direction. **E.g.** Light travels through vacuum.

Non-uniform velocity

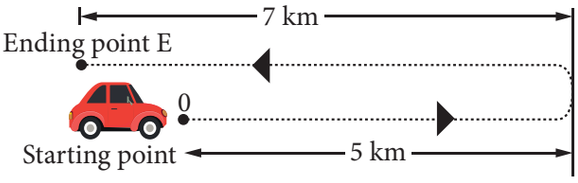
If either speed or direction changes, the velocity is non-uniform. **E.g.** A train starting and moving out of the station.

Average velocity

If the total displacement of an object is divided by the total time taken by the object we get the average velocity.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

In the figure given below, a car travels 5 km due east and makes a U – turn to travel another 7 km. If the time taken for the whole journey is 0.2 h, calculate the average velocity of the car.



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